

Chapter 5: A Survey of Probability Concepts

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Introduction to Probability

Consider a coin toss of a Bahraini coin. We know that there are only two possible **outcomes**: heads or tails.

Formally, we denote this as an experiment with two possible outcomes:

$$S = \{H, T\}$$

Introduction to Probability

Suppose we want to know the probability that the coin will show heads. We will formally denote this as **event** A :

$$A = \{\text{Coin shows heads}\} = \{H\}$$

We know that there are 2 possible states: heads or tails. Since only one of them can be shown, then the probability of event A occurring is simply:

$$P(A) = \frac{1}{2}$$

Introduction to Probability

Activity

We know that when we toss a coin, the possible outcomes are:

$$S = \{H, T\}$$

Determine the possible outcomes for the two experiments:

- (a) Rolling a six-sided die.
- (b) Tossing two coins.

Introduction to Probability

Solution:

(a) When you roll a die, there are only 6 possible outcomes:

$$S = \{1, 2, 3, 4, 5, 6\}$$

(b) When you toss two coins, there are 4 possible outcomes:

$$S = \{(H, T), (H, H), (T, H), (T, T)\}$$

Introduction to Probability

In general, when all possible outcomes are **equally likely**, the classical probability of an event A occurring is:

$$P(A) = \frac{\text{Number of outcomes in event } A}{\text{Total number of possible outcomes}}.$$

So if $S = \{H, T\}$ and event $A = \{H\}$, then $P(A)$ is equal to:

$$P(A) = \frac{1}{2}$$

Introduction to Probability

Activity:

Determine the following probabilities:

- (a) Rolling an even number from a six-sided die.
- (b) Tossing two coins and getting the same side.

Introduction to Probability

Solution:

(a) We know that for a six-sided die:

$$S = \{1, 2, 3, 4, 5, 6\}$$

which means we have a total of 6 possible outcomes. The event A is when the die reveals an even number:

$$A = \text{Even number from die roll} = \{2, 4, 6\}$$

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This is 3 of the possible 6 total outcomes. Therefore:

$$P(A) = \frac{3}{6} = \frac{1}{2}$$

Introduction to Probability

(b) When we toss two coins, we know that:

$$S = \{(H, T), (H, H), (T, H), (T, T)\}$$

which means we have a total of 4 possible outcomes. The event A is when both coins show the same side.

$$A = \text{Both coins show same side} = \{(H, H), (T, T)\}$$

Introduction to Probability

This is 2 of the possible 4 outcomes. Therefore:

$$P(A) = \frac{2}{4} = \frac{1}{2}$$

Approaches to Assigning Probabilities

When we are dealing with simple experiments, such as tossing a coin or rolling a die, it is relatively easy to list and count all the possible outcomes. However, as experiments become more complicated, listing every possible outcome individually can become difficult.

Approaches to Assigning Probabilities

In such cases, we can use different **counting methods** to determine the number of possible outcomes more efficiently. These methods can also help us determine the number of outcomes associated with a particular event.

The three counting methods we will consider are:

- **Multiplication**
- **Combinations**
- **Permutations**

Multiplication

Consider again the outcomes if you toss a coin. The coin can only give you heads H or tails T :

$$S_1 = \{H, T\}$$

Multiplication

Now suppose you toss 2 coins. The possible outcomes now become:

$$S_2 = \{(H, T), (H, H), (T, H), (T, T)\}$$

Multiplication

If you toss 3 coins, there possible outcomes become:

$$S_3 = \{(H, H, H), (H, H, T), (H, T, H), (H, T, T), \\ (T, H, H), (T, H, T), (T, T, H), (T, T, T)\}.$$

Multiplication

In general, you can determine the combinations by just multiplying the total outcomes of each coin. Since each coin can only be heads or tails:

- (a) One coin - 2 outcomes.
- (b) Two coins - $2 \cdot 2 = 4$ outcomes.
- (c) Three coins - $2 \cdot 2 \cdot 2 = 8$ outcomes.

Multiplication

Activity:

What are the total number of outcomes for the below experiments:

- (a) Tossing 4 coins.
- (b) Throwing 2 dices.

Multiplication

Solution:

(a) A coin can either be heads or tails. If a person tosses 4 coins, then:

$$2 \cdot 2 \cdot 2 \cdot 2 = 16 \text{ combinations}$$

(b) A die can have 6 outcomes. Therefore if a person throws two dices, the amount of combinations will be:

$$6 \cdot 6 = 36 \text{ combinations}$$

Combinations

A **combination** is used when we want to select a number of objects from a larger group and the **order of the selection does not matter**.

Consider a class with four students:

$$S = \{A, B, C, D\}.$$

A teacher needs to select 2 students to represent the class. They do this by randomly selecting any 2 from the 4 people in the classroom.

Combinations

The possible combinations of class representatives are:

$\{A, B\}$	$\{B, A\}$	$\{C, A\}$	$\{D, A\}$
$\{A, C\}$	$\{B, C\}$	$\{C, B\}$	$\{D, B\}$
$\{A, D\}$	$\{B, D\}$	$\{C, D\}$	$\{D, C\}$

A crucial point to see here is that the **order does not matter** here. For example, $\{A, B\} = \{B, A\}$.

Combinations

Rather than listing every possible committee, we can calculate the number of possible selections using the **combination rule**:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!},$$

where:

n = the total number of objects available,

r = the number of objects we want to select.

Combinations

In our example, there are four students available and we want to randomly select two of them. Therefore:

$$n = 4, \quad r = 2.$$

Applying the combination rule:

$$\binom{4}{2} = \frac{4!}{2!(4-2)!}$$

$$= \frac{4!}{2!2!}$$

$$= \frac{24}{2 \times 2} = 6.$$

Combinations

Therefore, there are **6 different committees** that can be formed by selecting two students from a group of four.

Permutations

A **permutation** is used when we want to select a number of objects from a larger group and the **order of the selection matters**.

Consider again a class with four students:

$$S = \{A, B, C, D\}.$$

Permutations

Suppose we want to select two students. However, this time the two students will have different positions: one student will be the **President** and the other will be the **Vice-President**.

For example, consider the following outcome:

(A, B) .

This means that student A is the President and student B is the Vice-President.

Permutations

Now consider:

(B, A) .

This represents a different outcome because student B is now the President and student A is the Vice-President.

Permutations

Therefore:

$$(A, B) \neq (B, A).$$

Unlike our previous committee example, changing the order of the students creates a different outcome. Therefore in this case, the **order matters**.

Permutations

We can list all the possible outcomes:

(A, B)	(B, A)	(C, A)	(D, A)
(A, C)	(B, C)	(C, B)	(D, B)
(A, D)	(B, D)	(C, D)	(D, C)

In each ordered pair, the first student is the President and the second student is the Vice-President. All 12 outcomes are therefore different.

Permutations

Rather than listing every possible outcome, we can calculate the number of possible arrangements using the **permutation rule**:

$$= \frac{n!}{(n - r)!}$$

where just as before:

n = the total number of objects available,

r = the number of objects being selected.

Permutations

In our example, there are four students available and two positions that need to be filled from selection:

$$n = 4, \quad r = 2.$$

Permutations

Applying the permutation rule:

$$= \frac{4!}{(4 - 2)!}$$

$$= \frac{4!}{2!}$$

$$= \frac{24}{2} = 12.$$

Permutations

Therefore, there are **12 different ways** of selecting a President and Vice-President from the four students.

Notation

To better understand this section, we need to introduce some notation from set theory, which is the concept of **union** and **intersection**.

Union

Consider a dice which has 6 sides. When rolling this dice, we should it to show one of the six possible outcomes:

$$S = \{1, 2, 3, 4, 5, 6\}.$$

all of which have an equally likely chance of occurring, which is $1/6$.

Union

Suppose we define two events from this dice roll A and B :

$$A = \text{An even number} = \{2, 4, 6\}$$

$$B = \text{A number greater than 3} = \{4, 5, 6\}$$

We are interested in the **union** of events A and B . This means we want to list out the numbers^a that result from combining A and B .

^aThis is called *elements* in set theory.

Union

We formally write this as:

$$A \cup B = \{2, 4, 5, 6\}$$

Notice how we do not double count the numbers. Meaning, even if 4 appears in both events, its union only includes it once.

We also generally refer the term union $\cup$ with the word **or**^b

^b $A \cup B = A \text{ or } B$

Intersection

Suppose now we want to know the **intersection** of events A and B . This means we want to list out the numbers are in both A and B . We formally write this as:

$$A \cap B = \{4, 6\}$$

Both 4 and 6 are numbers which are even and are also greater than 3. You can also easily see the overlapping numbers by looking at the numbers inside A and B .

Intersection

We generally refer the term intersection $\cap$ with the word **and**^c

We will use the union and intersection notation when analysing the probabilities of different events.

^c $A \cap B = A \text{ and } B$.

Rules of Addition

You should keep in mind two formulas which we will use which refer to the **rules of addition**.

Suppose we are interested in the **probability** that a number from A **or** B can occur. We calculate this by denoting:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B),$$

We will apply this formula when differentiating between events that are mutually exclusive and those that are not.

Mutually Exclusive Events

Let us begin by considering the case of **mutually exclusive events**. Two events are mutually exclusive if they cannot occur at the same time.

Consider again the experiment of rolling a fair six-sided die:

$$S = \{1, 2, 3, 4, 5, 6\}.$$

Mutually Exclusive Events

Suppose we define the following two events:

$A = \text{The die shows a } 2 = \{2\}$

$B = \text{The die shows a } 5 = \{5\}.$

When the die is rolled once, it cannot show both a 2 and a 5 at the same time. Therefore, events A and B are **mutually exclusive**.

Mutually Exclusive Events

Using the notation introduced earlier, we can see that the two events have no intersection:

$$A \cap B = \emptyset,$$

where $\emptyset$ represents an **empty set**. In other words, there are no outcomes that belong to both A and B and so $P(A \cap B) = 0$.

Mutually Exclusive Events

Suppose we are interested in the probability that the die shows a 2 **or** a 5. Using our notation for a union, we are interested in:

$$P(A \cup B).$$

Since the events are mutually exclusive, we can simply add their individual probabilities:

Mutually Exclusive Events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B)$$

In our example, we know that:

$$P(A) = \frac{1}{6}, \quad P(B) = \frac{1}{6}.$$

Mutually Exclusive Events

$$P(A \cup B) = P(A) + P(B)$$

$$= \frac{1}{6} + \frac{1}{6}$$

$$= \frac{2}{6}$$

Therefore, the probability that the die shows either a 2 or a 5 is $\frac{2}{6}$.

Non-Mutually Exclusive Events

Let us now consider the case of **non-mutually exclusive events**. Two events are non-mutually exclusive if they can occur at the same time. In other words, the two events have some outcomes in common.

Consider again the experiment of rolling a fair six-sided die:

$$S = \{1, 2, 3, 4, 5, 6\}.$$

Non-Mutually Exclusive Events

Suppose we define the following two events:

A = The die shows an even number = $\{2, 4, 6\}$

B = The die shows a number greater than 3 = $\{4, 5, 6\}$.

Notice that the outcomes 4 and 6 belong to both events. Therefore, events A and B are **non-mutually exclusive**.

Non-Mutually Exclusive Events

Using the notation introduced earlier, the intersection of the two events is:

$$A \cap B = \{4, 6\}.$$

Since there are two outcomes in the intersection, and each outcome has a probability of $1/6$, we have:

$$P(A \cap B) = P(A \text{ and } B)$$

$$= \frac{2}{6}$$

Non-Mutually Exclusive Events

Now suppose we are interested in the probability that the die shows an even number **or** a number greater than 3. Using our notation for a union, we are interested in:

Non-Mutually Exclusive Events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{3}{6} + \frac{3}{6} - \frac{2}{6}$$

$$= \frac{4}{6}$$

$$= \frac{2}{3}.$$

Therefore, the probability that the die shows an even number **or** a number greater than 3 is $\frac{2}{3}$.

The Rules of Multiplication

The rules of multiplication are used when we are interested in the probability of two or more events occurring together. However, we need to distinguish between **independent** and **dependent** events.

Independent Events

Two events are said to be **independent** when the occurrence of one event does not affect the probability of the other event occurring.

Consider an experiment where we roll two fair six-sided dice. Let us define the following two events:

A = The first die shows a 3

B = The second die shows a 6.

Independent Events

We know that the probability of each event is:

$$P(A) = \frac{1}{6}, \quad P(B) = \frac{1}{6}.$$

Events A and B are **independent**. The number shown by the first die does not affect the number that can be shown by the second die. For example, even if the first die shows a 3, the probability that the second die shows a 6 remains $1/6$.

Independent Events

When two events are independent, the probability that both events occur can be calculated by multiplying their individual probabilities:

$$P(A \cap B) = P(A) \times P(B).$$

Applying this rule to our example:

Independent Events

$$P(A \cap B) = P(A) \times P(B)$$

$$= \frac{1}{6} \times \frac{1}{6}$$

$$= \frac{1}{36}.$$

Therefore, the probability that the first die shows a 3 **and** the second die shows a 6 is $\frac{1}{36}$.

Dependent Events

Two events are said to be **dependent** when the occurrence of one event affects the probability of another event occurring.

Consider again the experiment of rolling a fair six-sided die:

$$S = \{1, 2, 3, 4, 5, 6\}.$$

Dependent Events

We define the following two events:

$A =$ The die shows a number greater than 3 $= \{4, 5, 6\}$

$B =$ The die shows an even number $= \{2, 4, 6\}$.

Dependent Events

Now suppose you throw the die. You are told by somebody that the number on the die is **greater than 3**, but they do not tell you which number. Basically, you know that event A has occurred, but you do not know which number it is.

Our possible outcomes are therefore restricted to:

$$\{4, 5, 6\}.$$

Dependent Events

After this has happened, you are now asked to determine the probability **that the die shows an even number**. That is, you need to determine the probability of event B **given that** event A has already occurred.

Formally, this is denoted as:

$$P(B|A) = \frac{2}{3}$$

Dependent Events

which is true, given that the only two numbers that are even is 4 and 6 out of the three possible outcomes.

Now let us step back to the point before the die has been rolled. We are interested in determining the probability of seeing an even number **given that** the number greater than 3. To do this, we apply the following multiplicative formula:

Dependent Events

$$P(A \cap B) = P(A) \cdot P(B|A)$$

Given that:

$$P(A) = \frac{3}{6}$$

$$P(B|A) = \frac{2}{3}$$

Dependent Events

therefore, the probability of this **dependent event** from occurring is:

$$P(A \cap B) = P(A) \cdot P(B|A)$$

$$= \frac{3}{6} \cdot \frac{2}{3}$$

$$= \frac{2}{6} = \frac{1}{3}$$

Contingency Table

The Sales Manager of Batelco is studying the patterns for placing international calls based on the age of their customers. They conduct a survey and collect the following results for September 2026:

Contingency Table

Table: International Calls by Customer Age

Customer	Age	Age Classification	International Calls
Customer 1	24	Young	2
Customer 2	42	Adult	1
Customer 3	67	Senior	0
Customer 4	31	Adult	3
Customer 5	19	Young	1
Customer 6	58	Adult	2
⋮			
Customer 500	72	Senior	1

Contingency Table

The results of the survey are then expressed in a contingency table:

Contingency Table

Table: International Calls by Age Classification

Calls Placed	Young	Adult	Senior	Total
0	15	50	10	75
1	25	100	75	200
2	55	60	60	175
3	5	15	30	50
Total	100	225	175	500

Contingency Table

Activity:

Determine the following probabilities:

- (a) The probability of a person placing 3 calls.
- (b) The probability of a person placing 1 call **or** less.
- (c) The probability of a person placing 3 calls **or** is a senior.
- (d) The probability of a person placing 3 calls **and** they are a senior.
- (e) The probability of a person placing 3 calls **given that** they are a senior.

Contingency Table

| **Solution:**

Contingency Table

(a) There are 50 people in the dataset who placed 3 calls out of a total of 500.

$$P(3 \text{ Calls}) = \frac{\text{Number of customers who placed 3 calls}}{\text{Total number of customers}}$$

$$= \frac{50}{500}$$

$$= 0.10.$$

Contingency Table

(b) We need to include the probability that the person either place 1 call **or** 0 calls.

$$P(1 \text{ Call or Less}) = P(0 \text{ Calls}) + P(1 \text{ Call})$$

$$= \frac{75}{500} + \frac{200}{500}$$

$$= \frac{275}{500}$$

$$= 0.55.$$

Contingency Table

(c) Remember that placing a call or being a senior are **not mutually exclusive**. That is, it is certainly possible for a person who placed 3 calls to also be a senior and so we need to subtract this intersection.

$$P(3 \text{ Calls or Senior}) = P(3 \text{ Calls}) + P(\text{Senior}) - P(3 \text{ Calls and Senior})$$

$$= \frac{50}{500} + \frac{175}{500} - \frac{30}{500}$$

$$= \frac{195}{500}$$

$$= 0.39.$$

Contingency Table

(d) We need to determine amongst all individuals, the probability that they placed 3 calls **and** they are a senior. We can see this intersection includes 30 people.

$$\begin{aligned}P(3 \text{ Calls and Senior}) &= \frac{\text{Number of seniors who placed 3 calls}}{\text{Total number of customers}} \\&= \frac{30}{500} \\&= 0.06.\end{aligned}$$

Contingency Table

(e) In this regard, "being a senior" is already an event that has occurred. So we are just trying to determine the probability that this senior placed 3 calls.

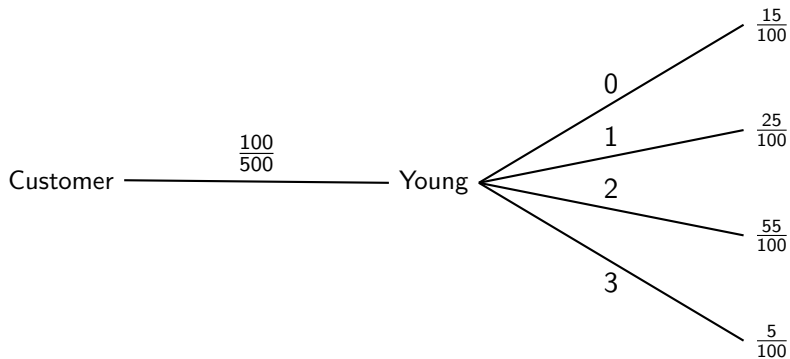
$$\begin{aligned}P(3 \text{ Calls} \mid \text{Senior}) &= \frac{\text{Number of seniors who placed 3 calls}}{\text{Total number of seniors}} \\&= \frac{30}{175} \\&= 0.171.\end{aligned}$$

Contingency Table

We can also visualise the probability outcomes using a Tree Diagram:

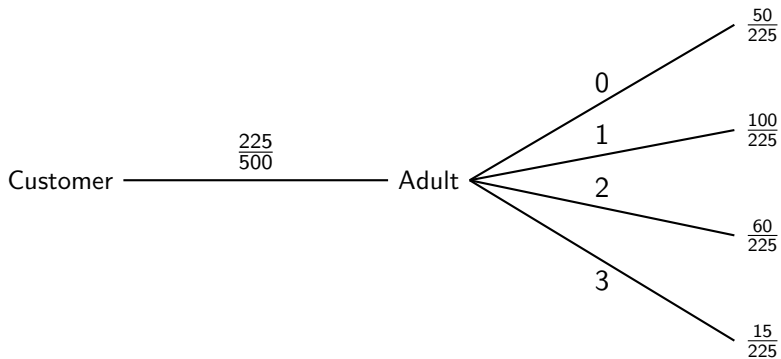
Contingency Table

Figure: Probability Tree for International Calls by Age Classification (Young)



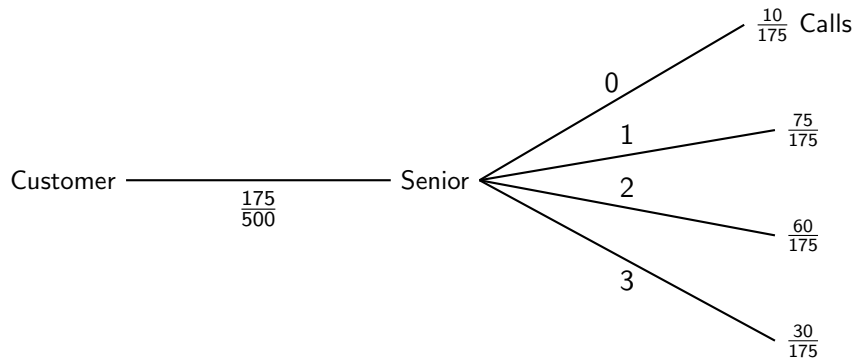
Contingency Table

Figure: Probability Tree for International Calls by Age Classification (Adult)



Contingency Table

Figure: Probability Tree for International Calls by Age Classification (Senior)



The Complement Rule

The **complement** of an event A consists of all outcomes where event A does **not** occur. We denote the complement of A as A^c .

Consider rolling a fair six-sided die:

$$S = \{1, 2, 3, 4, 5, 6\}.$$

The Complement Rule

Suppose we define event A as rolling an even number:

$$A = \{2, 4, 6\}.$$

The complement of A contains all the outcomes where A does not occur. Therefore:

The Complement Rule

$$A^c = \{1, 3, 5\}.$$

Since an event must either occur or not occur, the probabilities of an event and its complement must add to 1:

$$P(A) + P(A^c) = 1.$$

The Complement Rule

Therefore, the **Complement Rule** is:

$$P(A^c) = 1 - P(A).$$

In our example:

The Complement Rule

$$P(A^c) = 1 - P(A)$$

$$= 1 - \frac{3}{6}$$

$$= \frac{3}{6}$$

$$= \frac{1}{2}.$$

Therefore, the probability of **not** rolling an even number is $\frac{1}{2}$.